CHANDAN BARIK

Senior Software Engineer | Java · Spring Boot · Microservices · Backend Architecture
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PROFESSIONAL SUMMARY

Backend software engineer with 5+ years of experience designing, scaling, and optimizing Java and Spring Boot microservices across travel-tech, HR-tech, and retail POS domains. Proven record of cutting API response times by 30–60%, eliminating database and algorithmic bottlenecks, and replacing costly third-party dependencies with in-house solutions. Strong in REST and GraphQL API design, event-driven architecture with Apache Kafka, Spring Batch data pipelines, PostgreSQL/MongoDB modelling, and AWS deployment. Owns delivery end to end — from bottleneck analysis and low-level design through implementation, testing, deployment, and production validation. Oracle Certified Professional: Java SE 17 Developer.

TECHNICAL SKILLS

Languages: Java (8/11/17), SQL, C++

Frameworks: Spring Boot, Spring Data JPA, Spring Batch, MyBatis, Apache Camel

Architecture & APIs: Microservices, REST APIs, GraphQL, Event-driven design, ONDC protocol integration, Third-party API integration, Low-Level Design (LLD), High-Level Design (HLD)

Databases: PostgreSQL, MySQL, MongoDB, Vector databases, Semantic / vector search, Query optimization, Indexing strategies

Messaging & Streaming: Apache Kafka, AWS SQS, Asynchronous & parallel processing

Cloud & DevOps: AWS (EC2, S3, API Gateway, SQS), Docker, Git, CI/CD

Testing & Quality: TDD, Cucumber, Newman, Postman, Integration testing, Code reviews

Practices: System design, Performance tuning, Database optimization, Agile / Scrum, Production support & migrations

PROFESSIONAL EXPERIENCE

Publicis Groupe | Senior Associate Technology L1 (Senior Software Engineer)

Gurugram, India | Sep 2025 – Present

- Implemented a robust **asynchronous task processing system** in Java / Spring Boot using @Async executors, thread-pool tuning, and message-driven workers, reducing critical workflow completion time from **~8 minutes to under 30 seconds** and improving user experience.
- Redesigned high-cost queries, **indexing strategies**, and data-access patterns across core modules using query-plan analysis (EXPLAIN ANALYZE), composite indexes, and JPA/Hibernate query tuning, cutting average query execution time from **~5 minutes to ~50 seconds** and significantly reducing database load at peak traffic.
- Optimized time-critical algorithms and business workflows in Java by lowering computational complexity in hot code paths and eliminating redundant computation, reducing response time for high-volume operations from **~10 minutes to ~40 seconds**.
- Led the **migration of critical application components off expensive third-party utilities** to in-house Java / Spring Boot implementations and open-source alternatives, eliminating recurring licensing spend while improving maintainability and control over the technology stack.
- Drove independent **deep-dive investigations** across production systems using application profiling, query-plan analysis, and log/metric tracing, uncovering critical correctness, performance, and cost flaws in the database and application layers that had previously gone unnoticed.
- Proposed and **led the fixes end to end** — root-cause analysis, solution design, implementation, and production validation — converting one-off findings into lasting architectural improvements.
- Presented findings and remediation plans to engineering and product stakeholders, building consensus on priorities and driving the resulting changes through design and code reviews.

PeopleStrong | Software Development Engineer II (SDE II)

Gurugram, India | May 2025 – Sep 2025

- Reduced API response time by **30–60%** across legacy Java EE modules by profiling hot paths and removing redundant computation on a large multi-tenant HR platform.
- Refactored monolithic service logic to eliminate duplicate database calls and rewrite complex SQL/HQL queries operating on high-volume employee datasets.
- Introduced **asynchronous processing** for heavy-lift operations, freeing request threads and improving throughput and stability under peak load.

- Coordinated with QA and product teams to validate every optimization against existing functionality, shipping performance gains with zero functional regressions.

OutPath (Vaipratech Solutions) | Software Engineer — Java Backend
Remote / Chennai, India | Nov 2022 – May 2025

- Architected and led backend development of a multi-modal **travel planning and booking platform** built on Spring Boot microservices, automating journey creation and booking that was previously manual.
- Designed and delivered a **journey planner and metro ticketing service** integrated through **ONDC** (Open Network for Digital Commerce) protocols, extending the product into urban transit.
- Integrated multiple third-party Travel Service Provider APIs for **flights, buses, and cabs** (RedBus, TBO), building resilient REST clients with retries, fallbacks, and normalized response contracts.
- Implemented a **GraphQL** API layer that eliminated client-side over-fetching, cut payload sizes, and improved web and mobile response times.
- Built a real-time streaming pipeline with **Apache Kafka** and **AWS SQS** to process booking events, inventory updates, and status changes reliably at scale.
- Designed **Spring Batch** jobs for high-volume fare, inventory, and price synchronization, keeping catalogue data near real-time across suppliers.
- Added **vector / semantic search** over travel content to power natural-language destination and service discovery.
- **Tech:** Spring Boot, JPA, MyBatis, PostgreSQL, MongoDB, Kafka, GraphQL, AWS API Gateway, SQS, EC2, S3.

Toshiba India Pvt. Ltd. | Software Engineer (promoted from Software Engineer Intern)
Bengaluru, India | Jun 2021 – Nov 2022

- Developed backend services for an enterprise **Point of Sale (POS)** platform using Java, Spring Boot, Spring Batch, and MyBatis.
- Engineered and documented **RESTful APIs** connecting POS terminals with inventory management and financial/payment systems.
- Contributed to **high-level and low-level design** of a microservices-based POS architecture, improving modularity and reducing time-to-market for new features.
- Led adoption of automated testing using **Cucumber, Newman, and Postman**, reducing manual QA effort by **25%** and saving **100+ man-hours per month**.
- Achieved **90%+ code coverage** through unit and integration tests, significantly reducing post-deployment defects.
- Participated in agile ceremonies, code reviews, and production troubleshooting to improve system stability.

EDUCATION

B.Tech, Computer Science & Engineering
Chandigarh Engineering College, Mohali (Punjab Technical University) | 2018 – 2022 | CGPA: 8.9/10

Senior Secondary (CBSE), Science — PCM
NR Public School, Khurja | 2017 | 92%

CERTIFICATIONS & ACHIEVEMENTS

- **Oracle Certified Professional: Java SE 17 Developer** — Oracle (2026)
- **Machine Learning Essentials** — AWS | **Data Science Virtual Internship** — BCG
- **Google Digital Unlocked** (Digital Marketing Fundamentals) | **Google Web Designer Basics**
- **IIT Bombay** certification in HTML, PHP & MySQL | Qualified **LinkedIn C++ assessment**
- **CodeChef** rating 1635 | **170+** problems solved on LeetCode | **Smart India Hackathon** participant

LANGUAGES

English (Full Professional) | Hindi (Full Professional) | Odia (Working Proficiency)